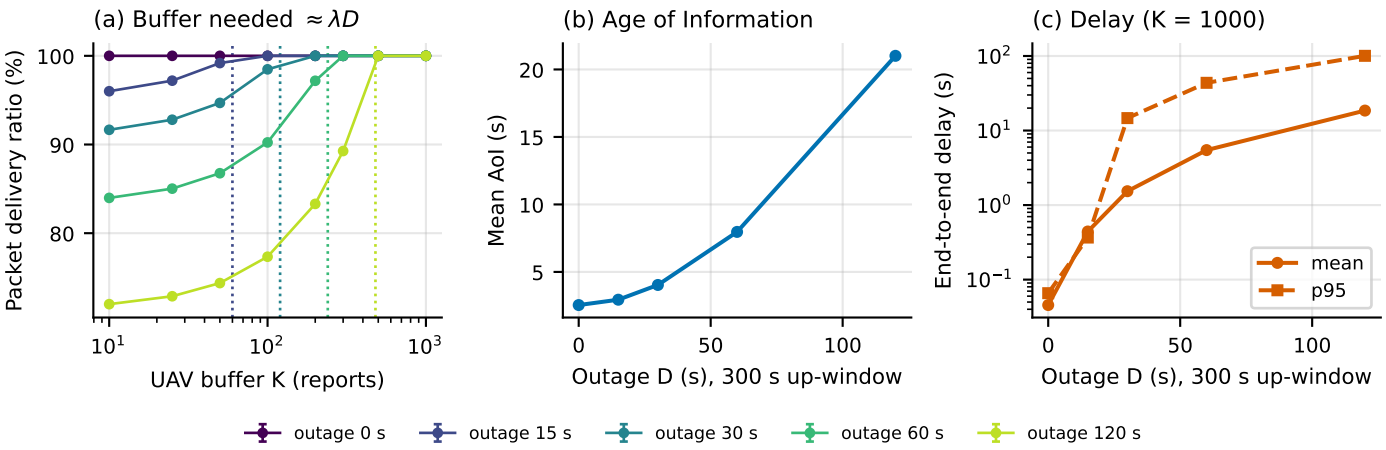
Store-and-forward under UAV link outages: SF7, 20 devices at 1 report/5 s (λ = 4 pkt/s)
(a) Buffer needed ≈ λD
Packet delivery ratio (%)
100
90
80
10^1
10^2
10^3
UAV buffer K (reports)
(b) Age of Information
Mean AoI (s)
20
15
10
5
0
50
100
Outage D (s), 300 s up-window
(c) Delay (K = 1000)
End-to-end delay (s)
10^2
10^1
10^0
10^{-1}
mean
p95
0
50
100
Outage D (s), 300 s up-window
outage 0 s
outage 15 s
outage 30 s
outage 60 s
outage 120 s